Chapter 7 — Communication Engineering

RRB JE CBT-2 · Electronics & Allied Engineering (CEN 05/2025, Annexure-VII C) · Syllabus Unit 7

📊 Weightage & Trend Box

Parameter	Detail
Approx. weightage	~16% of technical marks — one of the two heaviest units
Questions in 22-04-2025 CBT-2 paper	**8 out of 50** technical questions
Difficulty profile	Mostly conceptual one-liners + 1–2 direct formula numericals
Hottest sub-topics	PLL applications (FM & AM detection), pulse modulation family (PWM/PDM/PPM), digital modulation (QPSK), pulse shaping, AM power relations, Fourier symmetry

Actual questions observed in the 22-04-2025 paper (all solved in the PYQ Corner):

1. FM detection (demodulation) using a **PLL** — identifying the correct statement/output point.
2. Alternate name of **PWM** — Pulse **Duration** Modulation (PDM), a member of the Pulse **Time** Modulation (PTM) family.
3. **QPSK** — the four carrier phase shifts **0°, 90°, 180°, 270°**.
4. **Pulse-shaping filters** — identify the *non*-pulse-shaping filter among raised-cosine, sinc, Gaussian, and a distractor.
5. **AM sideband power**: 100% modulation, carrier 100 W ⇒ power in **LSB = 25 W**.
6. **Fourier transform of a real and even** function ⇒ result is **real and even**.
7. **AM detection using a PLL** — correct/incorrect statements (coherent detection, multiplier + LPF).
8. **PPM demodulation** — components used (ramp/pulse generator + **flip-flop** + LPF, or conversion to PWM first).

Preparation strategy: master (i) every block diagram (transmitter, superheterodyne receiver, PLL, PCM chain, PPM demodulator), (ii) the AM/FM power–bandwidth formula set, (iii) the pulse & digital modulation "family trees", and (iv) one-line facts on OFC, radar, cellular and satellite. That combination alone secures 6–8 marks.

7.1 Introduction to Communication

7.1.1 Elements of a Communication System

Every communication system, from a PA system in a railway station to a satellite link, has the same skeleton:

```
Information → Input      → Transmitter → Channel → Receiver → Output     → Destination
Source        Transducer                (+noise)              Transducer
```

Block	Function	Example
Information source	Produces the message (speech, music, picture, data)	Human voice
Input transducer	Converts message into an electrical signal (baseband signal)	Microphone, camera
Transmitter	Processes the baseband signal for efficient transmission — amplification, **modulation**, filtering	AM/FM broadcast transmitter
Channel	Physical medium carrying the signal; introduces attenuation, distortion and **noise**	Free space, coaxial cable, twisted pair, optical fibre
Receiver	Extracts (demodulates) the message from the received signal, amplifies it	Superheterodyne radio
Output transducer	Converts electrical signal back to the original message form	Loudspeaker, display

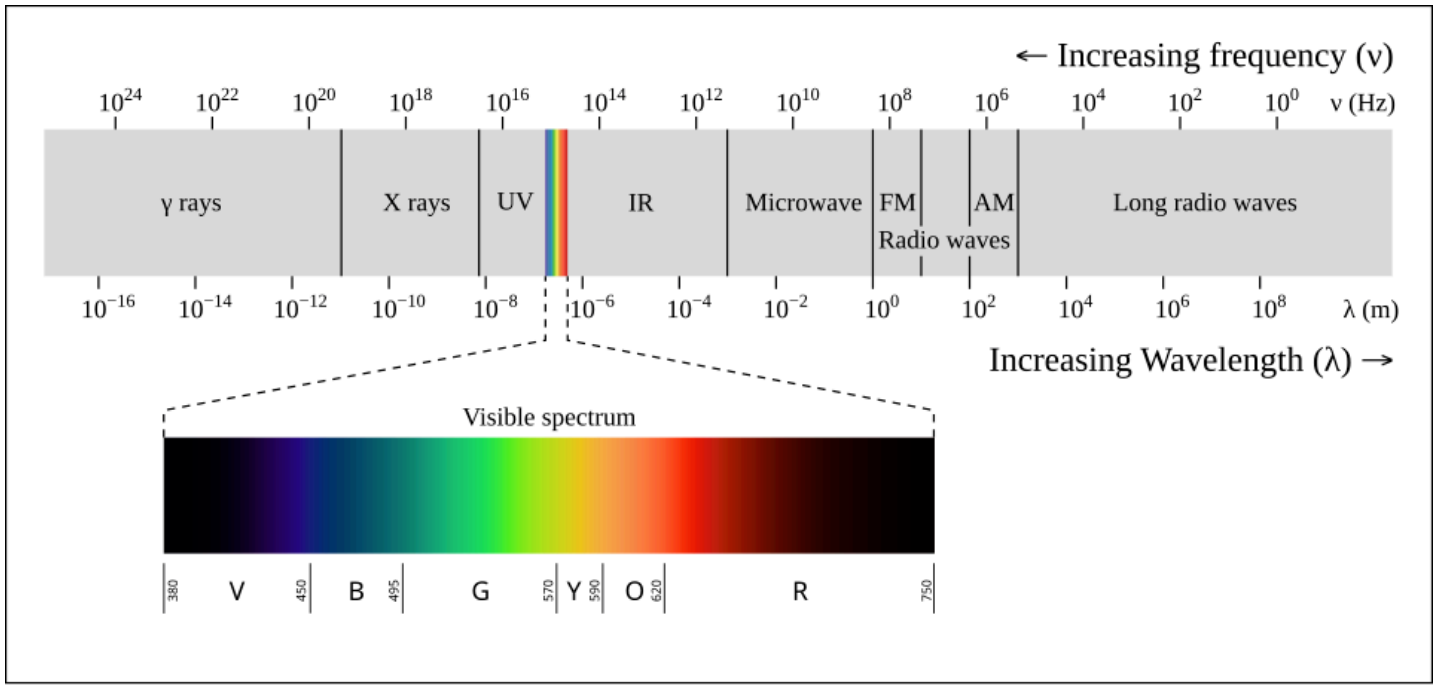
Baseband signal: the original band of frequencies produced by the source (speech ≈ 300 Hz–3.4 kHz; audio ≈ 20 Hz–20 kHz; video ≈ 0–4.2 MHz).

7.1.2 Why Modulate? (Need for Modulation)

Modulation = varying some parameter (amplitude, frequency, or phase) of a high-frequency **carrier** in accordance with the baseband (modulating) signal.

1. **Practical antenna length** — efficient radiation needs antenna ≈ λ/4. For 10 kHz audio, λ/4 = 7.5 km (impossible); for a 1 MHz carrier, λ/4 = 75 m (practical).
2. **Multiplexing** — many stations can share the channel by using different carrier frequencies (frequency translation avoids mutual interference).
3. **Reduced noise & interference** — FM particularly; also allows bandwidth–SNR trade-off.
4. **Range increase** — high-frequency signals propagate farther for a given power.
5. **Narrowbanding** — the ratio f_max/f_min of the transmitted band becomes ≈ 1, easing equipment design (antennas, amplifiers work over a narrow fractional bandwidth).

7.1.3 Frequency Bands (memorise — direct questions)



Electromagnetic spectrum.

Band	Frequency range	Typical use / propagation
VLF	3–30 kHz	Navigation, submarine comm (ground wave)
LF	30–300 kHz	Radio beacons (ground wave)
MF	300 kHz–3 MHz	**AM broadcast (535–1605 kHz)** — ground wave day, sky wave night
HF	3–30 MHz	Short-wave radio — **sky wave (ionosphere)**
VHF	30–300 MHz	**FM broadcast (88–108 MHz)**, TV — space wave (LOS)
UHF	300 MHz–3 GHz	TV, **cellular mobile**, GPS — LOS
SHF	3–30 GHz	**Satellite (C, Ku), radar, microwave links** — LOS
EHF	30–300 GHz	mm-wave, 5G high band, radio astronomy

7.1.4 Noise

Noise = unwanted random signal that corrupts the message.

External noise (enters from outside the receiver):

- **Atmospheric (static)** — lightning discharges; dominant below ~30 MHz.
- **Extraterrestrial** — solar noise, cosmic (galactic) noise (~8 MHz–1.4 GHz window).
- **Industrial (man-made)** — motors, ignition, arc welding; dominant in urban areas, 1–600 MHz.

Internal noise (generated inside equipment):

Type	Cause	Key facts
Thermal (Johnson) noise	Random motion of electrons in a resistor	$P_n = kTB$ (available power); $V_{n(rms)} = \sqrt{(4kTRB)}$. **White** noise — flat PSD.
Shot noise	Discrete arrival of charge carriers across a junction	$i_{n(rms)} = \sqrt{(2qI_{DC}B)}$
Partition noise	Random division of current between electrodes (BJT, mixer tubes)	Mixers noisier than amplifiers
Flicker (1/f) noise	Surface/contact imperfections	Dominant below ~1 kHz; PSD ∝ 1/f (pink noise)
Transit-time noise	Carrier transit time comparable to signal period	Significant at very high (microwave) frequencies

$k = 1.38 \times 10^{-23}$ J/K, T in kelvin, B = bandwidth in Hz.

Signal-to-noise ratio: SNR = S/N; in dB, $SNR(dB) = 10 \log_{10}(S/N)$.

Noise figure (F) — how much a device degrades SNR:

F = (SNR at input)/(SNR at output) ($F \geq 1$; ideal noiseless device $F = 1 \Rightarrow 0$ dB)

Noise figure in dB: $NF = 10 \log_{10} F$.

Equivalent noise temperature: $T_e = (F - 1)T_0$, with $T_0 = 290$ K.

Friis formula for cascaded stages (noise figure of a chain):

$F = F_1 + (F_2 - 1)/G_1 + (F_3 - 1)/(G_1G_2) + \ldots$

⇒ the **first stage** (LNA) dominates overall noise figure — put the low-noise, high-gain amplifier first.

7.2 Signals & Fourier Basics

7.2.1 Signal Classification

- **Analog vs digital**, **continuous-time vs discrete-time**.
- **Periodic**: $x(t) = x(t + T_0)$ — expandable in a **Fourier series** (discrete spectrum at nf_0).
- **Aperiodic / energy signals** — described by the **Fourier transform** (continuous spectrum).
- **Deterministic vs random** (noise is random — described statistically by PSD).

7.2.2 Fourier Transform

$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{(-j2\pi ft)}\, dt$ $x(t) = \int_{-\infty}^{\infty} X(f)\, e^{(+j2\pi ft)}\, df$

Standard pairs:

x(t)	X(f)
$\delta(t)$	1
1 (DC)	$\delta(f)$
$\cos 2\pi f_0 t$	$\frac{1}{2}[\delta(f-f_0) + \delta(f+f_0)]$
$\sin 2\pi f_0 t$	$(1/2j)[\delta(f-f_0) - \delta(f+f_0)]$
rect(t/T) (pulse of width T)	T·sinc(fT)
sinc(2Wt)	(1/2W)·rect(f/2W)
e^(−at)u(t), a>0	1/(a + j2πf)
Gaussian $e^{(-\pi t^2)}$	Gaussian $e^{(-\pi f^2)}$ (self-transform)

Key properties:

Property	Time domain	Frequency domain				
Linearity	$a\,x_1(t) + b\,x_2(t)$	$a\,X_1(f) + b\,X_2(f)$				
Time shift	$x(t - t_0)$	$X(f)\,e^{(-j2\pi f t_0)}$				
Frequency shift (modulation)	$x(t)\,e^{(j2\pi f_0 t)}$	$X(f - f_0)$				
Scaling	x(at)	(1/|	a|	) X(f/a)		
Duality	X(t)	x(−f)				
Convolution	$x_1(t) * x_2(t)$	$X_1(f)\cdot X_2(f)$				
Multiplication	$x_1(t)\cdot x_2(t)$	$X_1(f) * X_2(f)$				
Differentiation	dx/dt	$j2\pi f\,X(f)$				
Parseval	∫|	x(t)|	^{2}dt	= ∫|	X(f)|	^{2}df

7.2.3 Symmetry Properties — ⭐ Exam Favourite (asked 22-04-2025)

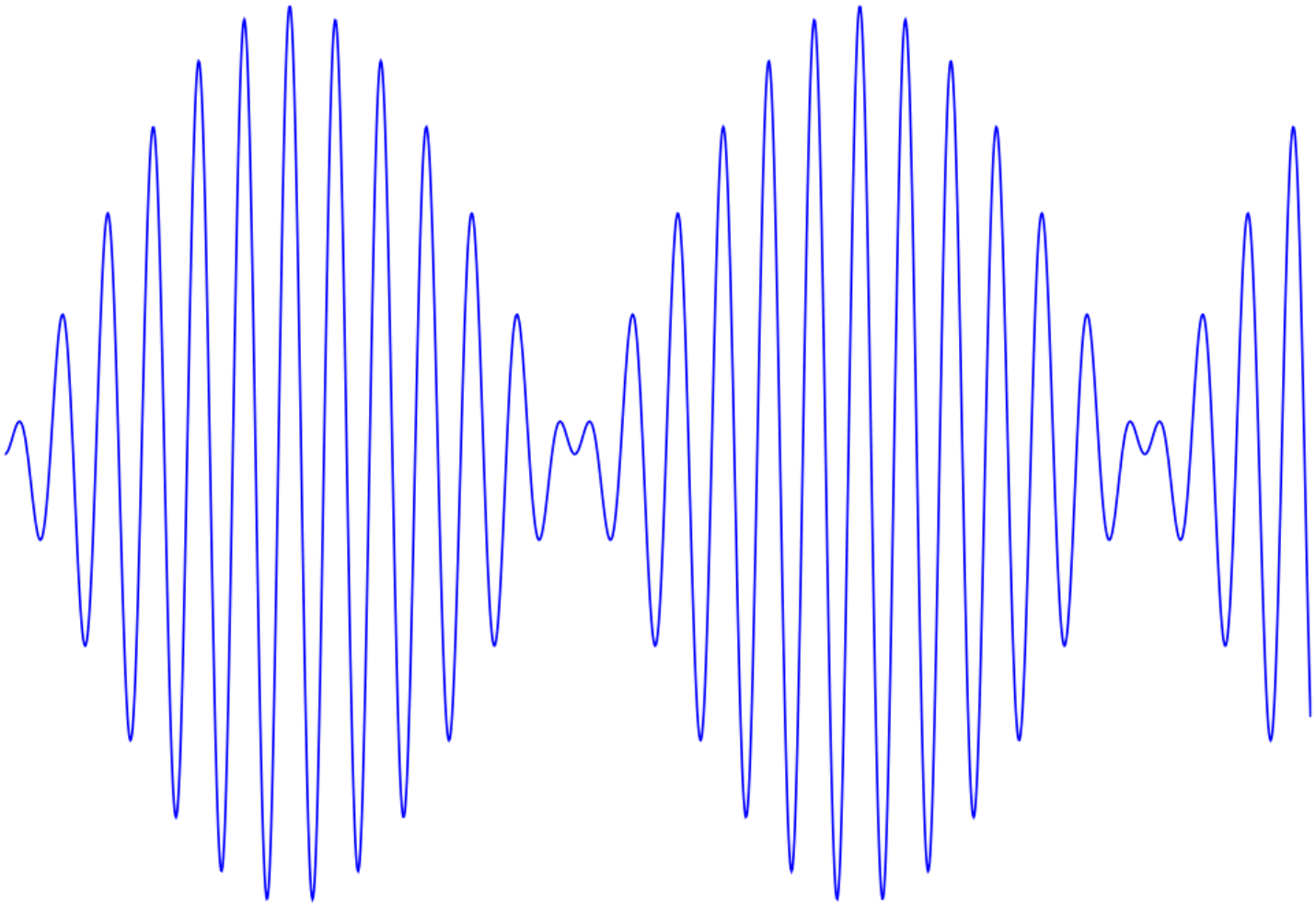
For x(t) real: X(−f) = X*(f) (**conjugate/Hermitian symmetry**) ⇒ |X(f)| is even, phase is odd.

If x(t) is …	Then X(f) is …
Real and even	**Real and even** ✅ (2025 PYQ)
Real and odd	**Imaginary and odd**
Imaginary and even	Imaginary and even
Imaginary and odd	Real and odd
Real (general)	Conjugate-symmetric: X(−f) = X*(f)

Memory hook: "*even keeps the character (real↔real), odd flips it (real↔imaginary).*"

7.3 Amplitude Modulation (AM)

7.3.1 Expression & Modulation Index



Amplitude modulation: message, carrier, AM wave.

Carrier: $c(t) = A_c \cos \omega_c t$. Message: $m(t) = A_m \cos \omega_m t$.

Standard AM (DSB-Full Carrier):

$s(t) = A_c[1 + m \cdot \cos \omega_m t] \cos \omega_c t$

Modulation index (depth):

$m = A_m/A_c = (V_{max} - V_{min})/(V_{max} + V_{min})$

where V_{max}, V_{min} are envelope maximum and minimum. Also $A_m = (V_{max} - V_{min})/2$, $A_c = (V_{max} + V_{min})/2$.

- $m < 1$: under-modulation; $m = 1$: 100% (critical) modulation; **$m > 1$: over-modulation** → envelope crosses zero → **envelope distortion & sideband splatter**; envelope detection fails.
- Percent modulation = $m \times 100\%$.
- **Multiple tones**: $m_t = \sqrt{(m_1^2 + m_2^2 + m_3^2 + \ldots)}$

7.3.2 Spectrum & Bandwidth

Expanding:

$s(t) = A_c \cos \omega_c t + (mA_c/2) \cos(\omega_c + \omega_m)t + (mA_c/2) \cos(\omega_c - \omega_m)t$

Three components: **carrier at f_c**, **USB at $f_c + f_m$**, **LSB at $f_c - f_m$**.

Bandwidth of AM: $B = 2f_m$ (twice the highest modulating frequency).

7.3.3 Power Relations — ☆ direct 2025 numerical

With carrier power $P_c = A_c^2/2R$:

$P_{total} = P_c(1 + m^2/2)$

Power per sideband: $P_{USB} = P_{LSB} = m^2P_c/4$ Total sideband power = $m^2P_c/2$.

Transmission efficiency: $\eta = P_{sidebands}/P_{total} = m^2/(2 + m^2)$.

- At $m = 1$: $\eta = 1/3 \approx$ **33.3%** (maximum for standard AM) — carrier wastes ≥ 2/3 of the power.

Current relation: $I_t = I_c\sqrt{(1 + m^2/2)}$ (antenna current).

> **2025 PYQ flash:** $P_c = 100$ W, $m = 1 \Rightarrow$ each sideband = $m^2P_c/4 = 100/4 =$ **25 W**; total = 150 W.

7.3.4 AM Variants — DSB-SC, SSB, VSB

Scheme	What is transmitted	Bandwidth	Power saving vs AM (m=1)	Detection	Uses
DSB-FC (standard AM)	Carrier + both sidebands	$2f_m$	—	Simple envelope detector	Broadcast radio
DSB-SC	Both sidebands, carrier suppressed	$2f_m$	66.7% (carrier removed)	Coherent (synchronous) only	Analog TV colour subcarrier, stereo pilot systems
SSB-SC	One sideband only	**f_m** (half)	83.3%	Coherent; needs precise carrier reinsertion	Point-to-point HF, marine, amateur radio
VSB	One full SB + vestige of other	$f_m < B < 2f_m$	intermediate	Envelope (with carrier)	**Analog TV video** (video has strong low frequencies; SSB filter impractical)

- DSB-SC generation: **balanced modulator / ring modulator** (multiplier).
- SSB generation: **filter method** (sharp BPF after balanced modulator), **phase-shift method** (two balanced modulators + 90° Hilbert phase shifters), **Weaver (third) method**.
- Coherent detection needs a locally generated carrier synchronised in **frequency and phase**; phase error θ scales output by $\cos \theta$ (quadrature null effect at $\theta = 90°$).

7.3.5 AM Generation

- **Low-level modulation**: modulate at low power then amplify with **linear** amplifiers.
- **High-level modulation**: modulate at the final high-power stage (collector/plate modulation) — allows efficient class-C carrier amplifiers.

- **Square-law modulator**: nonlinear device $i = a\,v + b\,v^2$; the v^2 term produces the product $m(t)\cdot c(t)$; a BPF at f_c selects the AM band. Works for small signals.
- **Switching modulator**: diode driven hard by the large carrier acts as a switch; output = $[c(t)+m(t)]\cdot g_p(t)$; BPF extracts AM.
- **Balanced modulator**: two matched square-law/switching modulators in push-pull cancel the carrier → DSB-SC.

7.3.6 AM Detection (Demodulation)

(a) Square-law detector — small-signal nonlinear device; output contains a term $\propto m(t)$ plus distortion (m^2 term); used only for very small signals.

(b) Envelope detector (the workhorse) — diode + parallel RC.

- Diode charges C on positive peaks; C discharges through R between peaks, tracking the envelope.
- **Design condition:** $1/f_c \ll RC \ll 1/f_m$ (more precisely $RC \le \sqrt{(1-m^2)}/(2\pi f_m \cdot m)$).
- **Diagonal (clipping) distortion**: RC too large → discharge slower than envelope fall → output follows a diagonal line instead of the envelope. Occurs when the envelope's negative slope exceeds the RC discharge slope; worst at high m and high f_m.
- **Negative-peak clipping**: caused by different DC/AC load impedances after the detector.

(c) Synchronous (coherent) detector — multiply received signal by a local carrier $\cos \omega_c t$, then low-pass filter:

$s(t)\cdot\cos \omega_c t \rightarrow \tfrac{1}{2}A_c[1+m\cdot\cos \omega_m t] +$ (terms at $2f_c$, removed by LPF)

Required for DSB-SC/SSB; also works for standard AM.

(d) PLL-based AM detection — ☆ 2025 PYQ.

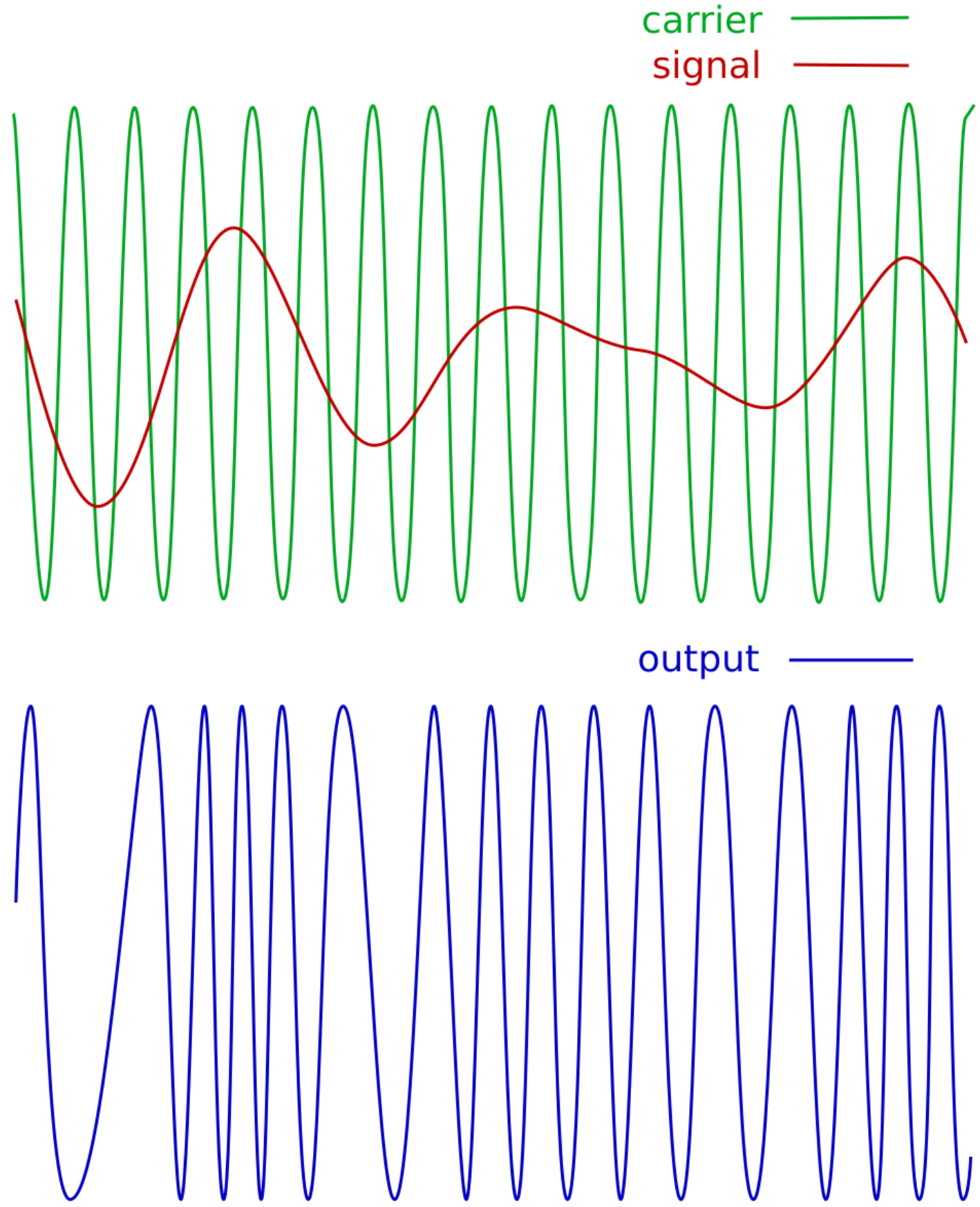
A PLL locks to the incoming AM carrier. Because the PLL's VCO output tracks the carrier **frequency and phase** but has **constant amplitude**, it serves as the clean, coherent local carrier. This VCO output (shifted 90° as needed so it is in phase with the incoming carrier) is fed to a **multiplier (product detector)** along with the AM input; the multiplier output through a **low-pass filter** yields the message m(t).

Correct statements to recognise:

- PLL regenerates a **coherent carrier** from the AM wave itself.
- Detection is **synchronous/coherent**, hence superior noise performance vs envelope detection and immune to selective carrier fading.
- The demodulated output is taken from the **multiplier + LPF**, *not* from the loop error voltage (contrast with FM, where the error/control voltage *is* the output).

7.4 Angle Modulation — FM & PM

7.4.1 Expressions



Frequency modulation waveform.

General angle-modulated wave: $s(t) = A_c \cos \theta(t)$, instantaneous frequency $f_i = (1/2\pi)\cdot d\theta/dt$.

Phase modulation (PM): $\theta(t) = \omega_c t + k_p m(t)$

$s(t) = A_c \cos[\omega_c t + k_p m(t)]$

Frequency modulation (FM): $f_i = f_c + k_f m(t)$

$s(t) = A_c \cos[\omega_c t + 2\pi k_f \int m(t)dt]$

For single-tone $m(t) = A_m \cos \omega_m t$:

$s_{FM}(t) = A_c \cos(\omega_c t + \beta \sin \omega_m t)$

Frequency deviation: $\Delta f = k_f A_m$ (maximum shift of instantaneous frequency from f_c).

FM modulation index: $\beta = \Delta f / f_m$ (dimensionless; can exceed 1 — unlike AM).

PM modulation index: $\beta_p = k_p A_m = \Delta\theta$ (peak phase deviation, independent of f_m).

FM–PM duality: PM of $\int m(t)dt$ ≡ FM of $m(t)$; FM of dm/dt ≡ PM of $m(t)$. (Integrator before a phase modulator ⇒ FM — Armstrong's idea.)

Deviation ratio (broadcast FM): $D = \Delta f_{max}/f_{m(max)} = 75\text{ kHz}/15\text{ kHz} = 5$.

7.4.2 Spectrum & Bandwidth

Single-tone FM expands in **Bessel functions**: $s(t) = A_c \Sigma J_n(\beta) \cos(\omega_c + n\omega_m)t$ — theoretically **infinite sidebands** at $f_c \pm nf_m$; amplitudes $J_n(\beta)$. Carrier amplitude $\propto J_0(\beta)$ and **vanishes** at $\beta \approx 2.405, 5.52, \ldots$ (used for deviation calibration).

Carson's rule (practical bandwidth ≈ 98% power):

$B = 2(\Delta f + f_m) = 2f_m(\beta + 1)$

- **NBFM**: $\beta \ll 1$ ($\beta < 0.3$): only first pair of sidebands significant; $B \approx 2f_m$ (like AM). Used in mobile/2-way radio.
- **WBFM**: $\beta \gg 1$: $B \approx 2\Delta f$. Commercial FM: $\Delta f = 75$ kHz, $f_m = 15$ kHz ⇒ $B = 2(75+15) =$ **180 kHz** (channel spacing 200 kHz).

7.4.3 FM Power

Amplitude is constant ⇒ **total power $= A_c^2/2R$ = carrier's unmodulated power, independent of β**. Modulation only redistributes power among sidebands. (Contrast AM: power grows with m.)

7.4.4 Noise in FM; Pre-emphasis & De-emphasis

- FM output noise PSD rises **parabolically with frequency** ($\propto f^2$) — high audio frequencies suffer more noise.
- **Pre-emphasis** (at transmitter): boost high frequencies before modulation — an RC high-pass/lead network, standard time constant **75 µs** (India/US; 50 µs Europe), corner ≈ 2.1 kHz.
- **De-emphasis** (at receiver): matching RC low-pass restores flat response while cutting the amplified high-frequency noise ⇒ net SNR improvement.
- FM also exhibits the **capture effect** (stronger signal suppresses the weaker co-channel one) and **threshold effect** (SNR advantage collapses below a carrier-to-noise threshold ≈ 10 dB).

7.4.5 FM Generation

Direct methods (vary oscillator frequency directly):

- **Varactor (voltage-variable capacitor) modulator** — message changes tank capacitance of an LC/VCO oscillator.
- **Reactance modulator** — FET/BJT circuit simulating a voltage-dependent reactance across the tank.

- Advantage: large deviation directly. Drawback: carrier (LC oscillator) has poor frequency stability — needs AFC against a crystal reference.

Indirect (Armstrong) method:

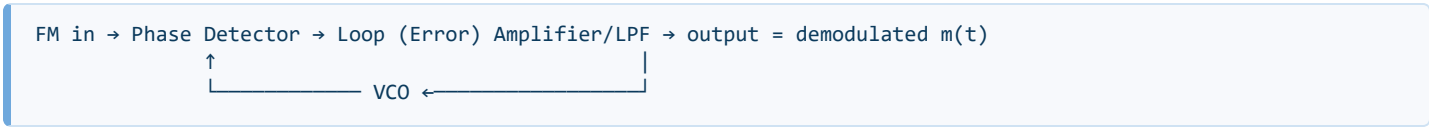
1. Crystal oscillator (stable carrier) → 90° phase shift.
2. Message → **integrator** → balanced modulator with the shifted carrier → adding back the carrier gives **NBFM** (small β).
3. **Frequency multipliers** (×n nonlinear stages) multiply both f_c and Δf; a mixer (heterodyne) repositions f_c without changing Δf.

⇒ Wideband FM with crystal stability. (Multiplier scales deviation; mixer does not.)

7.4.6 FM Detection

Detector	Principle	Notes
Slope detector	Off-tuned LC circuit converts Δf → amplitude variation, then envelope detector	Simple, nonlinear, poor
Balanced slope (Travis)	Two slope detectors tuned above/below f_c, outputs subtracted	Better linearity, hard to align (3 tuned circuits)
Foster–Seeley (phase) discriminator	Phase shift of a tuned coupled circuit varies with frequency; two diodes rectify vector sums	Very linear; **responds to amplitude variations too ⇒ needs a preceding limiter**
Ratio detector	Modified Foster–Seeley: one diode reversed + large stabilising capacitor holds sum voltage constant	**Built-in amplitude limiting** (≈ no separate limiter needed); output = half of discriminator
Quadrature detector	90° phase-shift network + phase comparator (IC receivers)	Common in TV sound ICs
PLL demodulator ☆	See below	Best linearity & threshold performance; no tuned circuits

PLL FM demodulator — ☆ 2025 PYQ (know this cold):



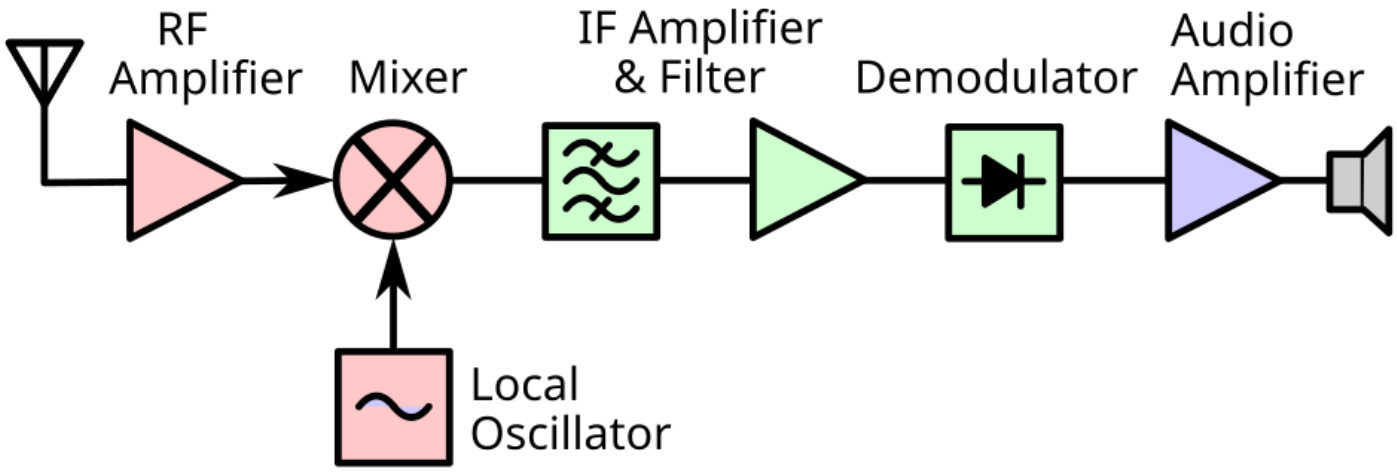
- The **phase detector** compares the incoming FM signal with the VCO output.
- The loop filter/**error amplifier** output (control voltage) drives the **VCO** so that VCO frequency tracks the instantaneous input frequency.
- In lock, VCO frequency $f_{VCO} = f_i(t) = f_c + k_f m(t)$. Since VCO frequency is proportional to its control voltage, the **control (error) voltage is directly proportional to m(t)** — *the loop-filter/error-amplifier output IS the demodulated signal.*
- Advantages: high linearity (set by VCO), inductor-less/IC-friendly, extends threshold, self-tuning.
- Contrast with PLL-AM detection: FM output = **loop error voltage**; AM output = **separate multiplier + LPF** using the VCO as coherent carrier.

7.4.7 AM vs FM (quick table)

Aspect	AM	FM
Parameter varied	Amplitude	Frequency
Bandwidth	2f_m (small)	2(Δf+f_m) (large)
Power	Varies with m	Constant
Noise immunity	Poor	Excellent (constant envelope + limiter)
Circuits	Simple	Complex
Coverage	Large (sky/ground wave, MF/HF)	Line-of-sight (VHF)
Broadcast band	535–1605 kHz	88–108 MHz

7.5 Radio Receivers

7.5.1 TRF vs Superheterodyne



Superheterodyne receiver block diagram.

TRF (Tuned Radio Frequency): several cascaded RF amplifiers all tuned to the station frequency → gain/selectivity vary across the band, tracking of multiple tuned stages is hard, unstable at high gain. Obsolete.

Superheterodyne (all modern receivers): convert every incoming station to a fixed **intermediate frequency (IF)** where most gain & selectivity are provided.

```
Antenna → RF amplifier → Mixer → IF amplifier(s) → Detector → AF amplifier → Speaker
                          ↑            (455 kHz AM / 10.7 MHz FM)
                  Local Oscillator (ganged tuning: f_LO = f_s + f_IF)
```

- **Standard IF: 455 kHz (AM broadcast), 10.7 MHz (FM), 38.9 MHz vision IF (TV, India)**.
- Local oscillator usually **above** the signal: f_LO = f_s + f_IF (smaller capacitor tuning ratio).
- **AGC** (automatic gain control): DC from detector controls IF/RF gain to keep output constant.

7.5.2 Image Frequency — classic numerical source

The mixer also converts f_image = **f_s + 2f_IF** (LO above) to the same IF — an unwanted "image" station.

- Image must be rejected **before** the mixer (RF stage selectivity); after mixing it is inseparable.
- **Image Frequency Rejection Ratio**: IFRR = $\sqrt{(1 + Q^2\rho^2)}$, where $\rho = f_{im}/f_s - f_s/f_{im}$ and Q = loaded Q of the RF tuned circuit. Cascaded stages: multiply IFRRs.
- Higher f_{IF} ⇒ image farther away ⇒ easier rejection (but harder selectivity); hence double-conversion receivers.

7.5.3 Receiver Characteristics

- **Sensitivity** — minimum input signal for a standard output (μV); set by gain & noise figure.
- **Selectivity** — ability to reject adjacent channels; mostly from IF filters.
- **Fidelity** — faithful reproduction of the whole baseband (needs adequate bandwidth — conflicts with selectivity).
- **Double spotting** — one station received at two dial points due to poor image rejection.

7.6 Pulse Modulation

7.6.1 Sampling Theorem — the gateway

Nyquist/Shannon sampling theorem: a band-limited signal with highest frequency f_m is completely recoverable from its samples if the sampling rate

$f_s \geq 2f_m$ (Nyquist rate = $2f_m$; Nyquist interval = $1/2f_m$)

- $f_s < 2f_m$ ⇒ **aliasing** (spectral overlap; irreversible distortion). Anti-aliasing LPF precedes the sampler.
- Types of sampling: **ideal (impulse)**, **natural** (pulse tops follow signal), **flat-top** (sample-and-hold; introduces **aperture effect**, corrected by a $(\sin x/x)^{-1}$ equaliser).
- Telephony: f_m = 3.4 kHz → standard **f_s = 8 kHz**.

7.6.2 Analog Pulse Modulation Family

Scheme	Parameter varied	Class	Bandwidth	Noise immunity
PAM	Pulse **amplitude**	—	Lowest of the three	Poor (amplitude carries info)
PWM = PDM = PLM	Pulse **width/duration/length**	**PTM**	Higher	Good
PPM	Pulse **position** (timing)	**PTM**	Highest	Best; also most power-efficient (constant-width, constant-amplitude narrow pulses)

☆ **2025 PYQ:** PWM is also called **Pulse Duration Modulation (PDM)** (and Pulse Length Modulation, PLM). PWM and PPM together form the **Pulse Time Modulation (PTM)** family because the information rides on pulse *timing*, not amplitude.

Generation:

- **PAM**: sample-and-hold / multiplying the message with a pulse train.
- **PWM**: compare message with a **ramp/sawtooth** in a comparator — the crossing instant sets the trailing edge ⇒ width ∝ message. (Also: monostable multivibrator with message-controlled timing.)
- **PPM from PWM**: differentiate PWM, keep the trailing-edge trigger, fire a **monostable** ⇒ constant-width pulse whose *position* ∝ message.

Detection:

- **PAM**: low-pass filter (plus equaliser for flat-top).
- **PWM**: simple **low-pass filter** (average value of PWM ∝ width ∝ message) — after regeneration/limiting.
- **PPM demodulation — ⭐ 2025 PYQ:** first convert PPM back to PWM, then LPF. Standard circuit: a **reference (synchronising) pulse generator + flip-flop (bistable)**: the reference clock pulse **sets** the flip-flop; the received PPM pulse **resets** it. The flip-flop output is a PWM wave whose width equals the position delay; a **low-pass filter** then recovers m(t). (Alternative: ramp generator started by the reference pulse and stopped by the PPM pulse — ramp height ∝ position.)

Components to tick in the MCQ: **pulse/ramp generator, flip-flop (bistable multivibrator), low-pass filter**.

7.6.3 PCM — Pulse Code Modulation (digital)

```
m(t) → LPF (anti-alias) → Sampler → Quantiser → Encoder → line coder → channel
Receiver: regenerator → decoder → reconstruction LPF
```

- **Quantisation**: rounding samples to L = 2^n discrete levels (n bits/sample). Step size Δ = (V_pp)/L.
- **Quantisation noise power** = $\Delta^2/12$ (uniform quantiser).
- **SQNR for full-scale sinusoid:**

SQNR = $1.5 \cdot L^2$ ⇒ SQNR(dB) = 6.02 n + 1.76 ≈ (6 dB per bit)

Each extra bit ⇒ **+6 dB** SNR but ×(n+1)/n bandwidth.

- **Bit rate**: R_b = n·f_s. **Minimum channel bandwidth** ≈ R_b/2 = n·f_s/2.
- **Companding** — non-uniform quantisation for speech: **μ-law (μ = 255, N. America/Japan)**, **A-law (A = 87.6, India/Europe)** — compress at Tx, expand at Rx to keep SNR uniform across amplitudes.
- Telephony PCM: 8 kHz × 8 bits = **64 kbps per voice channel** (DS0).
- Advantages: regeneration (noise does not accumulate), TDM-friendly, encryption/coding possible. Cost: bandwidth.

7.6.4 DPCM, DM, ADM

- **DPCM (Differential PCM)**: transmit the quantised **difference** between the sample and its predicted value — exploits correlation, fewer bits for the same quality.
- **Delta modulation (DM)**: 1-bit DPCM — staircase approximation, step ±Δ per clock.
- **Slope overload distortion** when Δ/T_s < |dm/dt|_max (signal rises faster than the staircase). Avoid: Δ f_s ≥ 2π f_m A_m for a tone.
- **Granular (idle) noise** when Δ too large for a slowly varying/flat signal.
- **ADM (Adaptive DM / CVSD)**: step size grows during overload, shrinks in granular regions — resolves the Δ conflict.

7.7 Digital (Carrier) Modulation

7.7.1 Binary Keying

Scheme	Carrier parameter keyed	Signal for bit 1 / 0	Bandwidth (≈, bit rate R_b)	Detection	Notes
ASK (OOK)	Amplitude	$A \cos \omega_c t$ / 0	$2R_b$	Envelope or coherent	Worst noise performance; simplest
FSK	Frequency	f_1 / f_2	$2(\Delta f + R_b)$ (Carson-like)	Non-coherent (discriminator) or coherent	Used in modems, RTTY; MSK = CPFSK with h=0.5
BPSK	Phase	0° / 180°	$2R_b$ (main lobe)	Coherent (Costas loop)	Best binary error performance
DPSK	Phase difference	data in phase *changes*	$2R_b$	Differentially coherent (no carrier recovery)	~1 dB worse than BPSK

Error-rate ranking (best → worst at same E_b/N_0): **BPSK/QPSK > DPSK > coherent FSK > non-coherent FSK > ASK**.

7.7.2 QPSK — ☆ 2025 PYQ

- 2 bits per symbol ⇒ 4 symbols ⇒ 4 carrier phases.
- **Conventional QPSK phases: 0°, 90°, 180°, 270°** (equivalently π/4, 3π/4, 5π/4, 7π/4 for offset constellation — but when the exam lists 0/90/180/270 vs other sets, tick **0°, 90°, 180°, 270°**).
- Symbol rate = $R_b/2$ ⇒ **bandwidth is half of BPSK** for the same bit rate, with the same error performance per bit.
- Implementation: serial-to-parallel splitter → I and Q BPSK modulators on $\cos \omega_c t$ and $\sin \omega_c t$ → adder. Variants: OQPSK (Q delayed T_b — avoids 180° transitions), π/4-QPSK.

Gray coding on constellations: adjacent symbols differ in one bit — minimises bit errors.

7.7.3 M-ary & QAM

- **M-PSK**: $\log_2 M$ bits/symbol on one amplitude circle; error rate worsens as points crowd.
- **QAM**: both **amplitude and phase** carry data (square constellations 16-QAM, 64-QAM, 256-QAM …). Best bandwidth efficiency; needs high SNR & linear amplifiers. Used in Wi-Fi, LTE/5G, cable modems, microwave links.
- Bandwidth efficiency $\eta = R_b/B$ (bits/s/Hz) increases with $\log_2 M$; power efficiency decreases.

7.7.4 Pulse Shaping & ISI — ☆ 2025 PYQ

Rectangular data pulses have sinc-shaped spectra with huge sidelobes; band-limiting them smears pulses into neighbours → **Inter-Symbol Interference (ISI)**.

Nyquist criterion for zero ISI: the overall pulse $p(t)$ must satisfy $p(kT_s) = 0$ for all integers $k \neq 0$ (zero crossings at all other sampling instants). Minimum bandwidth solution = **ideal sinc pulse** (brick-wall filter, $B = R_s/2$) — unrealisable and timing-sensitive.

Practical pulse-shaping filters:

Filter	Key property
Sinc (ideal Nyquist)	Minimum bandwidth $R_s/2$; unrealisable; slow $1/t$ tail decay
Raised-cosine (RC)	Roll-off factor α (0…1); $B = (1+\alpha)R_s/2$; zero-ISI zero crossings preserved; tails decay $\propto 1/t^3$
Root-raised-cosine (RRC)	RC split equally between Tx and Rx (matched filtering); cascade = RC
Gaussian	Used in **GMSK/GSM**; very compact spectrum, small controlled ISI; no zero-ISI property but excellent spectral containment

Exam trap (2025): raised-cosine, sinc (ideal Nyquist) and Gaussian **are** pulse-shaping filters; options like a plain **Butterworth/Chebyshev/high-pass/notch** filter are **not** pulse-shaping filters — that is the odd one out.

Eye diagram: oscilloscope overlay of symbol periods; eye opening ⇒ ISI margin & timing jitter tolerance.

7.8 Multiplexing Techniques

Multiplexing = combining several message signals for transmission over one channel.

Technique	Resource divided	Guard element	Typical use
FDM	Frequency slots (each signal SSB-translated to its own subcarrier)	**Guard bands**	Analog telephony groups, radio/TV broadcast, ADSL
TDM	Time slots (samples interleaved)	**Guard times**; needs frame **synchronisation**	Digital telephony (E1/T1), SONET/SDH
WDM/DWDM	Optical wavelengths on one fibre (FDM at light frequencies)	Channel spacing (e.g., 0.8 nm/100 GHz)	Fibre backbones
CDM/CDMA	Orthogonal/PN spreading codes — all users share full band & time	Code cross-correlation	3G (IS-95/WCDMA), GPS
SDM	Space (separate beams/antennas/MIMO)	Spatial isolation	Cellular sectoring, MIMO

FDM telephone hierarchy (classic): Group = 12 channels (48 kHz), Supergroup = 5 groups = 60 ch, Mastergroup = 5 supergroups = 300 ch.

TDM digital hierarchy — memorise the two primary rates:

Standard	Channels	Frame	Bit rate
T1 (DS1, N. America/Japan)	24 × 64 kbps + 8 kbps framing (193 bits/frame × 8000)	125 µs	**1.544 Mbps**
E1 (Europe/India)	32 slots × 64 kbps (30 voice + TS0 sync + TS16 signalling)	125 µs	**2.048 Mbps**

Higher orders: E2 = 8.448, E3 = 34.368, E4 = 139.264 Mbps; T2 = 6.312, T3 = 44.736 Mbps.

- **Synchronous vs statistical TDM**: fixed slots vs on-demand slots (better efficiency for bursty data).
- **OFDM** (used in 4G/5G/Wi-Fi/DVB): data split over many **orthogonal** closely-spaced subcarriers via IFFT; cyclic prefix combats multipath ISI.

7.9 Wave Propagation

7.9.1 Modes

Mode	Frequency range	Mechanism	Range
Ground (surface) wave	Up to ~2 MHz (VLF–MF)	Vertically polarised wave glides along earth; attenuation grows with f	Few hundred km; AM daytime
Sky wave (ionospheric)	**3–30 MHz (HF)**	Refraction (bending) by ionised layers	Thousands of km with hops; short-wave broadcast
Space wave (tropospheric/LOS)	> 30 MHz (VHF and above)	Direct + ground-reflected rays, line of sight	Tens of km; FM, TV, mobile, radar

Others: **tropospheric scatter** (100 MHz–10 GHz, beyond-horizon links), **duct propagation** (super-refraction layers).

7.9.2 Ionospheric Layers

Layer	Height (approx.)	Behaviour
D	50–90 km	Exists only in daytime; **absorbs** MF/HF (does not reflect usable HF); disappears at night → distant AM reception at night
E	90–140 km	Daytime reflection up to ~20 MHz; sporadic-E patches
F_1	140–250 km	Daytime only; merges into F at night
F_2	250–400 km	**Main reflecting layer**, day & night; highest electron density

7.9.3 Key Formulas

- **Critical frequency** (highest frequency returned at vertical incidence):

$f_c = 9\sqrt{N_{max}}$ (Hz, N_{max} = max electron density per m^3).

- **Maximum Usable Frequency** (oblique incidence at angle θ from vertical):

$MUF = f_c \cdot \sec\theta = f_c/\cos\theta$ (secant law). Optimum working frequency ≈ 0.85 MUF.

- **Skip distance**: minimum ground distance at which a sky wave of given frequency returns; between ground-wave limit and skip distance lies the **skip (silent) zone**.
- **Virtual height**: apparent reflection height assuming straight-line travel.
- **LOS (radio horizon) range** for antenna heights h_t, h_r (metres), with 4/3-earth correction:

$d\ (km) \approx 4.12(\sqrt{h_t} + \sqrt{h_r})$ (optical horizon: $3.57(\sqrt{h_t} + \sqrt{h_r})$)

- Ionospheric refraction: refractive index $n = \sqrt{(1 - 81N/f^2)}$.

7.10 Transmission Lines

7.10.1 Primary & Secondary Constants

Primary constants (per unit length): R (Ω/m), L (H/m), G (S/m), C (F/m).

Secondary constants:

- **Characteristic impedance:** $Z_0 = \sqrt{[(R + j\omega L)/(G + j\omega C)]}$
- **Lossless / high-frequency line: $Z_0 = \sqrt{(L/C)}$** (purely resistive) — the exam default.
- Common values: 50 Ω (RF/coax), 75 Ω (TV/CATV), 300 Ω (twin-lead), 600 Ω (open-wire telephone).
- **Propagation constant:** $\gamma = \alpha + j\beta = \sqrt{[(R+j\omega L)(G+j\omega C)]}$; α = attenuation (Np/m), β = phase constant = $2\pi/\lambda = \omega/v_p$.
- **Velocity:** $v_p = \omega/\beta = 1/\sqrt{(LC)}$ (lossless); velocity factor VF = v_p/c (≈ 0.66 for solid-PE coax).
- **Distortionless line (Heaviside condition): R/L = G/C** ⇒ α frequency-independent, v_p constant — achieved historically by **loading coils** (increasing L).

7.10.2 Reflection, VSWR & Impedance Transformation

- **Voltage reflection coefficient at the load:**

$\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$, $|\Gamma| \le 1$.

Matched ($Z_L = Z_0$): $\Gamma = 0$. Open circuit: $\Gamma = +1$. Short circuit: $\Gamma = -1$.

- **VSWR:**

$S = (1 + |\Gamma|)/(1 - |\Gamma|) = V_{max}/V_{min}$, $1 \le S \le \infty$. Also $|\Gamma| = (S-1)/(S+1)$.

For purely resistive loads: $S = Z_L/Z_0$ or Z_0/Z_L (whichever ≥ 1).

- **Reflected power fraction** = $|\Gamma|^2$; transmitted fraction = $1 - |\Gamma|^2$. **Return loss** = $-20 \log|\Gamma|$ dB.
- **Input impedance of a lossless line of length l:**

$Z_{in} = Z_0 \cdot (Z_L + jZ_0 \tan \beta l)/(Z_0 + jZ_L \tan \beta l)$

Special cases (memorise):

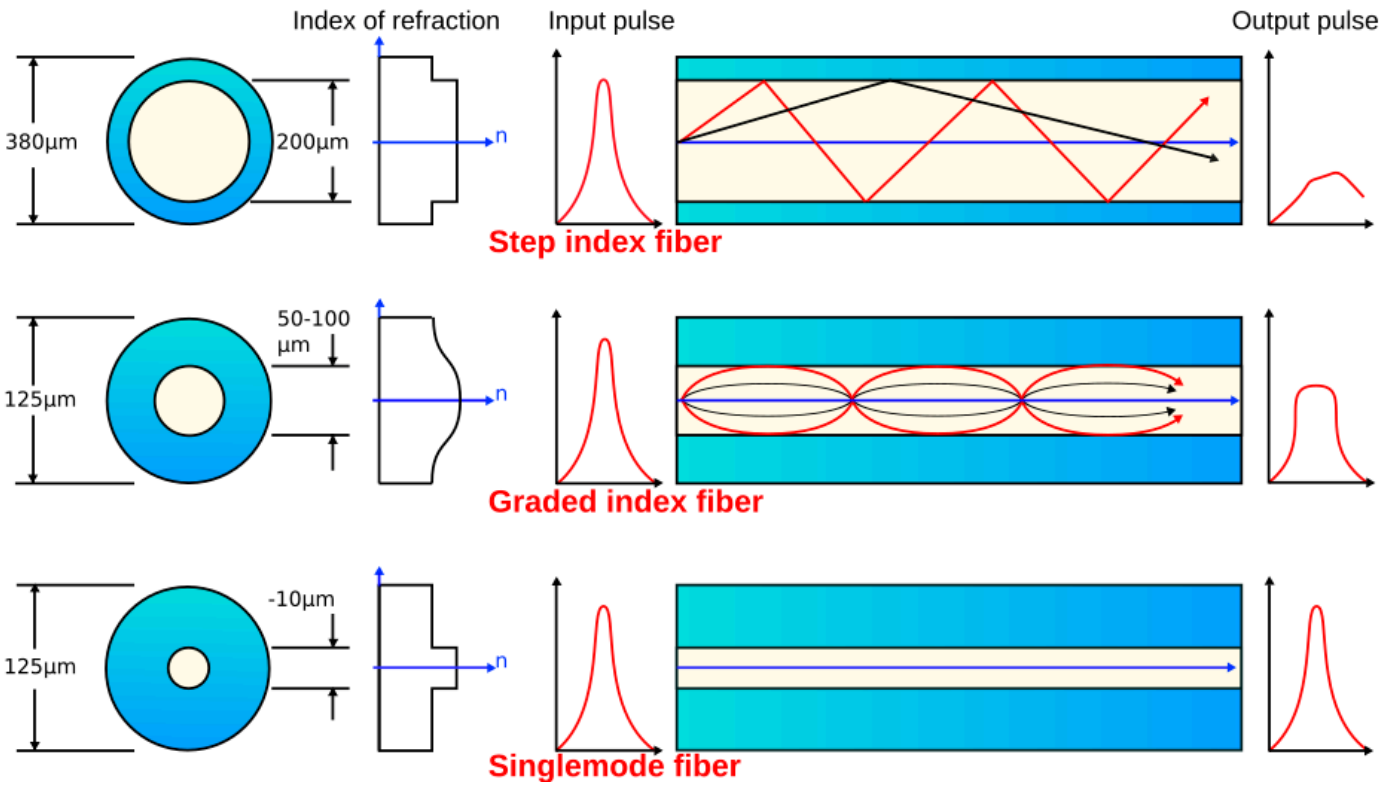
Line section	Z_in behaviour
λ/4 with load Z_L	**$Z_{in} = Z_0^2/Z_L$** (impedance inverter — quarter-wave transformer)
λ/2	$Z_{in} = Z_L$ (repeats every λ/2)
Shorted stub, l < λ/4	Inductive ($jZ_0 \tan \beta l$)
Open stub, l < λ/4	Capacitive ($-jZ_0 \cot \beta l$)
Shorted λ/4	Open circuit (parallel-resonant, "metallic insulator")
Open λ/4	Short circuit

7.10.3 Impedance Matching

- **Quarter-wave transformer**: to match real Z_L to Z_0, insert a λ/4 section of impedance **$Z_0' = \sqrt{(Z_0 \cdot Z_L)}$**. Narrow-band (length is λ/4 at one frequency).
- **Single-stub matching**: a short-circuited (preferred) or open stub in parallel at a computed distance from the load cancels the susceptance — done graphically on the **Smith chart**.
- **Double-stub**: two fixed-position stubs with adjustable lengths.

- **Smith chart**: polar plot of Γ; constant-r and constant-x circles; one full revolution = λ/2 along the line; centre = matched point.

7.11 Optical Fibre Communication (OFC)



Step-index vs graded-index optical fibre.

7.11.1 Structure & Guidance

Concentric layers: **core** (n_1, highest index) → **cladding** ($n_2 < n_1$) → buffer coating → strength members (Kevlar) → outer jacket. Light is guided by **total internal reflection (TIR)** at the core–cladding boundary (θ_incidence > critical angle θ_c = $\sin^{-1}(n_2/n_1)$).

7.11.2 Numerical Aperture & Acceptance Angle — numericals guaranteed

- **Numerical aperture: NA = $\sqrt{(n_1^2 - n_2^2)} = n_1\sqrt{(2\Delta)}$**, where **$\Delta = (n_1 - n_2)/n_1$** (fractional index difference; typically 0.01–0.03).
- **Acceptance angle** (half-angle of the entry cone, launch from air $n_0 = 1$):

$\theta_a = \sin^{-1}(NA)$; acceptance cone solid angle; rays inside the cone are guided.

- **Number of modes** (step-index): V number **$V = (2\pi a/\lambda)\cdot NA$**; modes ≈ $V^2/2$. **Single-mode condition: V < 2.405.**

7.11.3 Fibre Types

Type	Core dia.	Index profile	Dispersion	Use
Multimode step-index	50–200 µm	Abrupt $n_1 \rightarrow n_2$	Large **intermodal** dispersion	Short links, illumination
Multimode graded-index	50/62.5 µm	Parabolic (n max at axis)	Intermodal dispersion largely equalised (outer rays travel faster)	LANs, campus links
Single-mode step-index	**8–10 µm**	Abrupt, tiny core	No intermodal dispersion; only chromatic	**Long-haul, highest bandwidth**

7.11.4 Losses & Dispersion

- **Attenuation mechanisms**: material **absorption** (OH^- water peaks; intrinsic IR/UV), **Rayleigh scattering** ($\propto 1/\lambda^4$ — dominant at short λ), bending losses (macro/micro), splice/connector losses.
- **Low-loss windows**: **850 nm (~2–3 dB/km), 1310 nm (~0.35 dB/km, zero-dispersion), 1550 nm (~0.2 dB/km, minimum loss — long-haul/DWDM band)**.
- Loss in dB: $\alpha(dB) = 10 \log_{10}(P_{in}/P_{out})$; per-km figure = α/L.
- **Dispersion** (pulse spreading limits bit rate): intermodal (multimode), **chromatic** (material + waveguide; zero near 1310 nm in standard SMF), polarisation-mode dispersion (PMD).

7.11.5 Sources & Detectors

Parameter	LED	LASER diode
Emission	Spontaneous, incoherent	Stimulated, coherent
Spectral width	Broad (30–60 nm)	Narrow (< 1–2 nm)
Coupled power / speed	Low, ≤ ~200 Mbps	High, multi-Gbps
Linearity, cost, drive	Simple, cheap, rugged	Threshold current, temperature control, costly
Fibre pairing	Multimode, short links	Single-mode, long-haul

Detectors: **PIN photodiode** (simple, no gain) and **APD — avalanche photodiode** (internal gain via avalanche multiplication, better sensitivity, higher bias & noise).

7.11.6 Advantages of OFC (list the exam loves)

Enormous bandwidth; lowest loss (≈ 0.2 dB/km) ⇒ long repeater spacing; complete **immunity to EMI** & crosstalk; no ground loops, sparkless (safe in hazardous areas); light weight, small size; high security (no radiation, hard to tap); raw material (silica) abundant. Limitations: splicing/termination skill, bend sensitivity, opto-electronic conversion cost.

7.12 Public Address (PA) Systems

A PA system = **microphones → mixer/pre-amplifier → (tone/graph EQ) → power amplifier → loudspeakers**, reinforcing sound over a large area (railway platforms, auditoria).

7.12.1 Microphones

Type	Principle	Impedance	Notes
Carbon	Resistance of carbon granules varies with pressure	Low	Old telephones; needs DC; noisy
Moving-coil (dynamic)	Coil in magnetic field — electromagnetic induction	Low (~200–600 Ω)	Rugged, no external power — the PA workhorse
Ribbon (velocity)	Thin metal ribbon in field	Very low	Bidirectional (figure-8), delicate
Condenser (capacitor)	Capacitance varies with diaphragm motion	Very high (needs FET buffer + phantom 48 V)	Best fidelity; studio
Electret	Permanently charged condenser	High	Cheap, small (phones, lapel mics)
Crystal (piezoelectric)	Piezoelectric voltage	Very high	High output, poor fidelity

Directional patterns: omnidirectional, cardioid (uni-), bidirectional. **Feedback (howling)** is controlled by cardioid mics, speaker placement, and equalisers.

7.12.2 Amplifiers & Loudspeakers

- Chain: pre-amplifier (raises mV mic signal, mixing, tone control) → driver → **power amplifier** (class B/AB push-pull or class D for efficiency).
- **Loudspeaker**: moving-coil cone driver; voice-coil impedance typically **4, 8 or 16 Ω**; woofer/mid/tweeter split by **crossover networks**; horn (compression) drivers for high-efficiency outdoor PA.
- **Impedance matching** — maximum power transfer from amplifier to speakers; mismatch causes power loss or amplifier damage.

7.12.3 Constant-Voltage (70 V / 100 V) Line — ☆ PA speciality

For **many speakers over long cable runs** (platforms, factories), low-impedance (8 Ω) distribution would need impractically thick cable. Instead:

- A **step-up transformer** at the amplifier raises the output to a standard line voltage — **70.7 V (US) or 100 V (India/Europe)** at full rated power.
- Each speaker has its own **step-down matching transformer** with power taps (e.g., 10 W, 5 W, 2.5 W); speakers connect in **parallel** across the line.
- High voltage ⇒ low current ⇒ **small I^2R cable loss**, easy addition/removal of speakers; total tapped power must not exceed amplifier rating.
- Effective load impedance seen by amplifier: $Z = V_{line}^2/P$ (e.g., 100 V line, 100 W ⇒ 100 Ω).

7.13 Telephony & Electronic Exchange

7.13.1 Basics

- Telephone set: microphone + receiver + **hybrid (2-wire↔4-wire)** + dialler + ringer; local loop = twisted pair to the exchange; exchange supplies **−48 V DC** battery.
- Signalling: **pulse (rotary) dialling** vs **DTMF** (dual-tone multi-frequency — each digit = one low-group + one high-group tone).
- Call progress tones: dial tone, ring-back, busy.
- **Erlang**: unit of telephone traffic (1 Erlang = one circuit occupied continuously); grade of service = blocking probability.

7.13.2 Switching Evolution → Electronic Exchange

Generation	Technology
Manual	Operator + plugboard
Strowger (step-by-step)	Electromechanical, direct pulse-driven selectors
Crossbar	Electromechanical matrix + common control
SPC electronic exchange	**Stored Program Control** — a computer (software) controls switching; e.g., E-10B, EWSD, OCB-283, Indian **C-DOT** exchanges
Digital SPC	**PCM voice + time-division switching**

SPC advantages: flexible services (call waiting, forwarding, conference), easy tariff/number changes by software, self-diagnostics, faster call set-up, detailed billing.

Digital switching fabrics: **Space switch (S)** — crosspoint matrix connects physical lines; **Time switch (T)** — Time-Slot Interchanger (TSI) writes PCM slots into memory and reads them in a different order; large exchanges combine them: **T-S-T** or S-T-S architectures.

7.13.3 PSTN Hierarchy

Subscriber → **Local exchange (end office)** → **Tandem/junction exchange** (intra-city) → **Trunk (toll) exchange** (inter-city, STD) → **International gateway (ISD)**. Modern networks flatten this with digital trunks and IP (NGN/VoIP), but the hierarchy remains the exam answer.

7.14 Radar

RADAR = RAdio Detection And Ranging. Transmit EM pulses; receive echoes from targets; measure delay (range), direction (antenna bearing) and Doppler shift (velocity).

7.14.1 Range Measurement

$R = c\cdot\Delta t/2$ — factor 2 because the pulse travels out **and** back ($c = 3 \times 10^8$ m/s). Handy: 1 µs delay ⇔ 150 m range.

- **Maximum unambiguous range**: $R_{unamb} = c/(2\cdot PRF)$ — echo must return before the next pulse (PRF = pulse repetition frequency).
- **Range resolution**: $\Delta R = c\tau/2$ (τ = pulse width) — shorter pulses resolve closer targets.
- **Duty cycle** = $\tau\cdot PRF$; average power = peak power × duty cycle.

7.14.2 Radar Range Equation

Received echo power:

$$P_r = (P_t\cdot G^2\cdot\lambda^2\cdot\sigma)/((4\pi)^3\cdot R^4)$$

⇒ maximum range:

$$R_{max} = [\ P_t\cdot G^2\cdot\lambda^2\cdot\sigma\ /\ ((4\pi)^3\cdot S_{min})\ ]^{(1/4)}$$

- σ = radar cross-section (m^2), S_{min} = minimum detectable signal, G = antenna gain (same antenna Tx/Rx via **duplexer** — TR/ATR switch).
- **$R_{max} \propto P_t^{(1/4)}$**: to double range, transmit power must rise **16×**.

7.14.3 Radar Types

Type	Principle	Use
Pulse radar	Timed pulses → range & bearing	Air surveillance, ATC
CW radar	Continuous wave; measures only **Doppler shift** $f_d = 2v_r/\lambda$ (no range)	Speed guns
FM-CW radar	Frequency-swept CW; beat frequency gives range **and** velocity	Radio altimeters, automotive radar
MTI / pulse-Doppler	Doppler filtering rejects stationary clutter	Weather, military
Secondary radar (SSR)	Interrogator + transponder replies	ATC identification (IFF)

Displays: **PPI (Plan Position Indicator)** — polar map; A-scope — amplitude vs range. Radar bands: L (1–2 GHz), S (2–4), C (4–8), **X (8–12, most common)**, Ku, Ka.

Doppler shift: $f_d = 2v_r/\lambda = 2v_r f/c$ (approaching ⇒ $+f_d$).

Applications: air-traffic control, weather monitoring, speed enforcement, navigation (ship/air), altimetry, remote sensing, defence.

7.15 Cellular Communication

7.15.1 Cell Concept & Frequency Reuse

- Service area divided into **hexagonal cells**, each served by a low-power base station — the same frequencies are **reused** in sufficiently separated cells ⇒ enormous capacity from limited spectrum.
- **Cluster**: group of N cells using the whole allocated spectrum once; valid $N = i^2 + ij + j^2$ = 1, 3, 4, **7**, 12, 19 … (N = 7 classic).
- **Frequency reuse factor = 1/N**; channels per cell = total/N.
- **Co-channel reuse ratio**: $D/R = \sqrt{(3N)}$ (D = distance between co-channel cells, R = cell radius). Larger N ⇒ less co-channel interference but lower capacity.
- Capacity expansion: **cell splitting** (smaller cells), **sectoring** (120°/60° directional antennas), microcells/femtocells.

7.15.2 Handoff (Handover) & Roaming

- **Handoff**: transferring an ongoing call to a new base station as the mobile crosses cell boundaries — decided on received signal strength with hysteresis margin.
- **Hard handoff** ("break-before-make", FDMA/TDMA-GSM) vs **soft handoff** ("make-before-break", CDMA — mobile connected to ≥ 2 base stations simultaneously).
- **Roaming**: obtaining service outside the home network. Core entities (GSM): BTS → BSC → MSC, with HLR (home register), VLR (visitor register), AuC, EIR; SIM holds IMSI.
- GSM 900: uplink 890–915 MHz, downlink 935–960 MHz, 200 kHz carriers, 8 TDMA slots; GMSK modulation.

7.15.3 Generations — one-line evolution (memorise)

Gen	Era	Technology	Signature
1G	1980s	Analog FM, FDMA (AMPS/NMT)	Voice only
2G	1990s	Digital — **GSM (TDMA)** / IS-95 (CDMA)	Voice + **SMS**; 2.5G GPRS/EDGE data
3G	2000s	**WCDMA/UMTS, CDMA2000**	Mobile data ~Mbps, video calls
4G	2010s	**LTE — OFDMA + MIMO, all-IP**	Mobile broadband 100 Mbps–1 Gbps
5G	2020s	**NR — massive MIMO, mm-wave, network slicing**	eMBB, URLLC (~1 ms), mMTC/IoT

7.16 Satellite Communication

7.16.1 Orbits & the Geostationary Satellite

- **Geostationary orbit (GEO)**: circular, **equatorial**, eastward, period = 23 h 56 min 4 s (sidereal day) ⇒ satellite appears fixed. **Height ≈ 35,786 km** above the equator (orbital radius ≈ 42,164 km); velocity ≈ 3.07 km/s.
- **Three GEO satellites** spaced 120° cover almost the whole earth (Clarke, 1945) — polar regions excluded.
- **Propagation delay**: one hop up+down ≈ 2 × 36,000 km/c ≈ **240–280 ms** — noticeable in voice.
- Other orbits: **LEO** (500–2000 km — Iridium, Starlink; low delay, many satellites), **MEO** (~20,200 km — **GPS**), **Molniya** (highly elliptical, high-latitude coverage). India: INSAT/GSAT (GEO), NavIC/IRNSS (GEO+GSO navigation).

7.16.2 Space Segment: Transponder

Transponder = the satellite's receive–amplify–retransmit chain:

Receive antenna → LNA → **frequency down-converter (uplink→downlink)** → channel filter → high-power amplifier (**TWTA**/SSPA) → transmit antenna.

- Typical C-band transponder bandwidth **36 MHz**; a satellite carries 24+ transponders.
- **Uplink frequency > downlink frequency** (uplink power is easier to raise on the ground; lower downlink frequency suffers less path loss/rain for the power-limited satellite).

Band	Uplink/Downlink (GHz)	Notes
C band	**6 / 4** (5.925–6.425 up, 3.7–4.2 down)	Rain-resistant; large dishes
Ku band	**14 / 12** (14–14.5 up, 11.7–12.2 down)	DTH TV, VSAT; rain fade significant
Ka band	30 / 20	High-throughput satellites; worst rain fade
X band	8 / 7	Military

- Bus subsystems: power (solar array + batteries), attitude & orbit control (station-keeping within ±0.05°), TT&C (telemetry, tracking & command), thermal.
- Earth station: large dish + feed, LNA/LNB, up/down converters, HPA; **EIRP = $P_t \cdot G_t$**; link budget uses free-space path loss FSPL(dB) = 92.4 + 20 log f(GHz) + 20 log d(km). G/T = figure of merit of receive station.

7.16.3 VSAT & Applications

- **VSAT (Very Small Aperture Terminal)**: small earth station (dish **0.75–2.4 m**), Ku/C band, usually a **star network through a central hub** (or mesh); access by TDMA/SCPC/DAMA.
- Applications: banking/ATMs, reservations (railways!), rural telephony & internet, DTH broadcast, disaster-recovery links, SCADA telemetry.
- Satellite services: FSS (fixed), BSS (broadcast/DTH), MSS (mobile — INMARSAT), navigation (GPS/NavIC), remote sensing, weather (INSAT-3DR).

7.17 Worked Numericals (12)

N1. AM power (the 2025 style). A 100 W carrier is amplitude-modulated to 100% by a sine tone. Find total power, power per sideband and efficiency.

$P_t = P_c(1 + m^2/2) = 100(1 + 0.5) =$ **150 W**.

Each sideband $= m^2P_c/4 = 100/4 =$ **25 W** (so both sidebands = 50 W).

$\eta = m^2/(2+m^2) = 1/3 \approx$ **33.3%**.

N2. Modulation index from antenna current. Unmodulated antenna current 10 A rises to 11.2 A with sinusoidal modulation. Find m.

$I_t = I_c\sqrt{(1+m^2/2)} \Rightarrow (11.2/10)^2 = 1 + m^2/2 \Rightarrow 1.2544 - 1 = m^2/2 \Rightarrow m^2 = 0.5088 \Rightarrow$ **$m \approx 0.713$ (71.3%)**.

N3. Multi-tone AM. P_c = 400 W modulated simultaneously by tones with $m_1 = 0.3$ and $m_2 = 0.4$. Total power?

$m_t = \sqrt{(0.09+0.16)} = 0.5$. $P_t = 400(1 + 0.25/2) = 400 \times 1.125 =$ **450 W**.

N4. Envelope readings. An AM display shows V_{max} = 20 V, V_{min} = 4 V. Find m, A_c, A_m.

$m = (20-4)/(20+4) = 16/24 =$ **0.667**; $A_c = (20+4)/2 =$ **12 V**; $A_m = (20-4)/2 =$ **8 V**.

N5. Carson's rule. An FM signal has deviation 50 kHz with a 10 kHz tone. Bandwidth and β?

$\beta = 50/10 =$ **5**. $B = 2(\Delta f + f_m) = 2(50+10) =$ **120 kHz**.

N6. FM deviation scaling. In FM, k_f gives Δf = 20 kHz for A_m = 2 V at f_m = 4 kHz. If amplitude doubles and frequency halves, find new β.

$\Delta f \propto A_m \Rightarrow \Delta f' = 40$ kHz; $f_m' = 2$ kHz $\Rightarrow \beta' = 40/2 =$ **20** (old β was 5 — β quadruples).

N7. Image frequency. An AM receiver (IF = 455 kHz, LO above signal) is tuned to 1000 kHz. Find LO and image frequencies.

$f_{LO} = 1000 + 455 =$ **1455 kHz**; $f_{image} = f_s + 2f_{IF} = 1000 + 910 =$ **1910 kHz**.

N8. Sampling & PCM bit rate. A 4 kHz voice signal is sampled at 25% above Nyquist and encoded with 8 bits/sample. Find f_s, bit rate and minimum bandwidth.

Nyquist = 8 kHz; $f_s = 1.25 \times 8$ k = **10 kHz**. $R_b = 10$ k $\times 8 =$ **80 kbps**. $B_{min} = R_b/2 =$ **40 kHz**.

N9. PCM SNR. How many bits per sample guarantee SQNR ≥ 40 dB, and what SQNR results?

$6.02n + 1.76 \geq 40 \Rightarrow n \geq 6.35 \Rightarrow$ **n = 7 bits**, giving $6.02\times7 + 1.76 \approx$ **43.9 dB**.

N10. Radar range. A radar echo returns 200 μs after transmission. Target range? Also the maximum unambiguous range for PRF = 1 kHz.

$R = c\Delta t/2 = (3\times10^8 \times 200\times10^{-6})/2 =$ **30 km**.

$R_{unamb} = c/(2 \cdot PRF) = 3\times10^8/(2\times1000) =$ **150 km**.

N11. Numerical aperture. A fibre has $n_1 = 1.50$ (core), $n_2 = 1.47$ (cladding). Find NA and the acceptance angle in air.

$NA = \sqrt{(1.50^2 - 1.47^2)} = \sqrt{(2.25 - 2.1609)} = \sqrt{0.0891} \approx$ **0.298**.

$\theta_a = \sin^{-1}(0.298) \approx$ **17.4°** (acceptance cone ≈ 34.7° full angle).

(Also $\Delta = (1.50-1.47)/1.50 = 0.02$; check: $NA = n_1\sqrt{(2\Delta)} = 1.5\sqrt{0.04} = 0.3$ ✓)

N12. VSWR & reflection. A 50 Ω line is terminated in 75 Ω. Find Γ, VSWR and % power reflected.

$\Gamma = (75-50)/(75+50) = 25/125 =$ **0.2**.

$S = (1+0.2)/(1-0.2) =$ **1.5** (= 75/50 ✓).

Reflected power $= |\Gamma|^2 = 0.04 \Rightarrow$ **4%** reflected, 96% delivered.

Bonus N13. Quarter-wave match. Match a 100 Ω antenna to a 50 Ω line with a λ/4 transformer.

$Z_0' = \sqrt{(50\times100)} = \sqrt{5000} \approx$ **70.7 Ω** (hence 70–75 Ω sections are common).

Bonus N14. Delta modulation slope overload. DM with $f_s = 64$ kHz, step $\Delta = 0.1$ V on a tone $A \sin(2\pi\times1000t)$. Max A without slope overload?

Need $\Delta \cdot f_s \geq 2\pi f_m \cdot A \Rightarrow A \leq (0.1 \times 64000)/(2\pi\times1000) = 6400/6283 \approx$ **1.02 V**.

7.18 PYQ Corner — 22-04-2025 CBT-2 (fully solved)

Q1. In the detection (demodulation) of an FM signal using a PLL, the demodulated output is obtained from —

(a) the VSO free-running port (b) the phase-detector RF port directly (c) **the error amplifier / loop-filter output (VCO control voltage)** (d) a separate envelope detector

Answer: (c). In lock the VCO tracks the instantaneous input frequency $f_c + k_f m(t)$. Because VCO frequency is proportional to its control voltage, that control (error) voltage is a replica of m(t). The PLL FM demodulator therefore delivers the message at the loop-filter/error-amplifier output — no tuned circuits, excellent linearity.

Q2. Pulse Width Modulation (PWM) is also known as —

(a) Pulse Amplitude Modulation (b) **Pulse Duration Modulation (PDM)** (c) Pulse Code Modulation (d) Delta Modulation

Answer: (b). PWM ≡ **PDM** ≡ Pulse Length Modulation (PLM). Together with PPM it belongs to the **Pulse Time Modulation (PTM)** family, since the information is carried by pulse timing (width/position), not amplitude.

Q3. In QPSK, the phase of the carrier takes which set of values?

(a) 0°, 45°, 90°, 135° (b) **0°, 90°, 180°, 270°** (c) 0° and 180° only (d) 0°, 120°, 240°, 360°

Answer: (b). QPSK maps 2 bits/symbol onto 4 equally spaced carrier phases separated by 90°: **0°, 90°, 180°, 270°** (the offset constellation 45°/135°/225°/315° is the equivalent rotated version, but against these options the standard answer is 0/90/180/270). Symbol rate = $R_b/2$, so QPSK needs half the bandwidth of BPSK at the same bit rate.

Q4. Which of the following is NOT a pulse-shaping filter?

(a) Raised-cosine filter (b) Sinc (ideal Nyquist) filter (c) Gaussian filter (d) **Notch/Butterworth band-stop type filter (the non-Nyquist distractor)**

Answer: (d). Pulse-shaping filters are designed to limit bandwidth while controlling ISI: the **sinc** (ideal Nyquist, minimum bandwidth), the **raised-cosine / root-raised-cosine** (practical zero-ISI family with roll-off α) and the **Gaussian** (GMSK/GSM — compact spectrum). A generic frequency-selective filter (notch, plain Butterworth band-stop, high-pass etc.) is not a pulse-shaping filter.

Q5. A 100 W carrier is amplitude-modulated to a depth of 100%. The power in the lower sideband is —

(a) 50 W (b) **25 W** (c) 100 W (d) 12.5 W

Answer: (b). $P_{LSB} = m^2 P_c/4 = (1)^2(100)/4 =$ **25 W**. (USB also 25 W; total transmitted power = 100(1 + ½) = 150 W; sidebands together carry 50 W = 33.3%.)

Q6. If x(t) is a real and even function of time, its Fourier transform X(f) is —

(a) imaginary and odd (b) **real and even** (c) complex with odd magnitude (d) imaginary and even

Answer: (b). For real x(t), $X(-f) = X^*(f)$. Adding even symmetry forces the imaginary (sine-transform) part to vanish: $X(f) = 2\int_0^{\infty} x(t) \cos 2\pi ft \, dt$ — a **real and even** function of f. (Real-odd ⇒ imaginary-odd; the "even preserves, odd swaps real↔imaginary" rule.)

Q7. Which statement about AM detection using a PLL is correct?

(a) The demodulated AM output is the PLL error voltage itself

(b) **The PLL regenerates a coherent carrier (VCO output locked to the incoming carrier); this carrier multiplies the AM signal and a low-pass filter recovers the message — i.e., synchronous detection**

(c) A PLL cannot be used for AM because AM has no frequency information

(d) The VCO amplitude follows the AM envelope

Answer: (b). The PLL locks to the AM carrier and its constant-amplitude VCO output acts as the clean local carrier (phase-aligned via a 90° shifter). Feeding this and the AM wave to a **product detector (multiplier) + LPF** performs coherent detection — better than envelope detection under noise and carrier fading. Note the contrast with FM: in FM the error voltage is the output; in AM the multiplier+LPF output is.

Q8. The demodulation of PPM typically employs —

(a) only an envelope detector

(b) **a reference pulse generator with a flip-flop (converting PPM to PWM) followed by a low-pass filter**

(c) a frequency discriminator

(d) a Schmitt trigger alone

Answer: (b). Standard PPM demodulator: a synchronised **reference pulse sets a flip-flop**; the arriving **PPM pulse resets it**. The flip-flop output is a PWM signal whose width equals the transmitted position shift; averaging it in a **low-pass filter** recovers m(t). (Equivalent ramp-generator version: ramp started by the reference pulse, sampled/stopped by the PPM pulse — height ∝ position.)

7.19 Practice MCQs (30)

1. The main reason for modulation is —

(a) to reduce noise (b) reduction of antenna height to practical size (c) increase message power (d) reduce bandwidth

2. Thermal noise power available from a resistor is —

(a) kT/B (b) kTB (c) kT^2B (d) independent of bandwidth

3. In a cascade of amplifiers, the overall noise figure is dominated by —

(a) last stage (b) first stage (c) middle stage (d) all equally

4. An AM wave has V_max = 12 V and V_min = 4 V. The modulation index is —

(a) 0.33 (b) 0.5 (c) 0.66 (d) 0.75

5. The maximum transmission efficiency of standard AM is —

(a) 25% (b) 33.3% (c) 50% (d) 100%

6. The bandwidth of a standard AM signal modulated by 5 kHz audio is —

(a) 5 kHz (b) 10 kHz (c) 2.5 kHz (d) 15 kHz

7. Analog TV video is transmitted using —

(a) SSB (b) DSB-SC (c) VSB (d) FM

8. Diagonal clipping in an envelope detector occurs when —

(a) RC too small (b) RC too large (c) m = 0 (d) carrier absent

9. A carrier is frequency-modulated by a 2 kHz tone producing 10 kHz deviation. The modulation index is —

(a) 0.2 (b) 2 (c) 5 (d) 20

10. For commercial FM (Δf = 75 kHz, f_m = 15 kHz), Carson bandwidth is —

(a) 90 kHz (b) 150 kHz (c) 180 kHz (d) 200 kHz

11. Total power of an FM wave when modulation index doubles —

(a) doubles (b) quadruples (c) halves (d) remains unchanged

12. Pre-emphasis in FM broadcasting boosts —

(a) low frequencies at Tx (b) high frequencies at Tx (c) high frequencies at Rx (d) carrier amplitude

13. The FM detector with built-in amplitude limiting is the —

(a) slope detector (b) Foster–Seeley discriminator (c) ratio detector (d) square-law detector

14. The Armstrong method generates FM using —

(a) varactor diode (b) reactance FET (c) phase modulation of an integrated message with a crystal carrier (d) PLL

15. A superheterodyne AM receiver tuned to 700 kHz (IF = 455 kHz, LO high) has image frequency —

(a) 245 kHz (b) 1155 kHz (c) 1610 kHz (d) 910 kHz

16. Standard IF of an FM broadcast receiver is —

(a) 455 kHz (b) 1.6 MHz (c) 10.7 MHz (d) 38.9 MHz

17. A signal band-limited to 6 kHz must be sampled at least at —

(a) 6 kHz (b) 3 kHz (c) 12 kHz (d) 24 kHz

18. In PPM, the message information is carried by the pulse —

(a) amplitude (b) width (c) position (d) shape

19. Adding one bit per sample to a PCM system improves SQNR by about —

(a) 3 dB (b) 6 dB (c) 1.76 dB (d) 12 dB

20. Slope-overload distortion in delta modulation is reduced by —

(a) decreasing step size (b) increasing step size or sampling rate (c) reducing sampling rate (d) removing the integrator

21. For the same bit rate, the bandwidth of QPSK compared with BPSK is —

(a) same (b) double (c) half (d) quadruple

22. GMSK, used in GSM, employs which pulse-shaping filter?

(a) raised cosine (b) sinc (c) Gaussian (d) Chebyshev

23. The E1 PCM primary multiplex rate is —

(a) 1.544 Mbps (b) 2.048 Mbps (c) 8.448 Mbps (d) 64 kbps

24. Long-distance HF (short-wave) communication uses —

(a) ground wave (b) sky wave (c) space wave (d) duct propagation

25. If maximum ionospheric electron density is 1.21×10^{12} per m^3, the critical frequency is about —

(a) 4.95 MHz (b) 9.9 MHz (c) 11 MHz (d) 1.21 MHz

26. A lossless line has L = 0.5 µH/m and C = 200 pF/m. Its characteristic impedance is —

(a) 25 Ω (b) 50 Ω (c) 75 Ω (d) 100 Ω

27. A quarter-wave lossless line shorted at the far end appears at its input as —

(a) short circuit (b) open circuit (c) pure inductance (d) matched load

28. A fibre with NA = 0.242 launched from air has acceptance angle —

(a) 7° (b) 14° (c) 24.2° (d) 28°

29. The height of a geostationary satellite above the earth's surface is approximately —

(a) 6,378 km (b) 20,200 km (c) 35,786 km (d) 42,164 km

30. In a cellular system with cluster size N = 7, the co-channel reuse ratio D/R is —

(a) 3 (b) $\sqrt{21} \approx 4.58$ (c) 7 (d) 21

Answer Key & Brief Explanations

#	Ans	Explanation
1	b	Antenna ≈ λ/4; only practical at high (carrier) frequency.
2	b	$P_n = kTB$ — proportional to bandwidth.
3	b	Friis: later stages divided by preceding gains — first stage (LNA) dominates.
4	b	$m = (12-4)/(12+4) = 8/16 = 0.5$.
5	b	$\eta_{max} = m^2/(2+m^2)$ at $m = 1 \Rightarrow 1/3 = 33.3\%$.
6	b	$B_{AM} = 2f_m = 10$ kHz.
7	c	Vestigial sideband — saves bandwidth yet keeps low-frequency video content.
8	b	Large RC cannot discharge fast enough to follow the falling envelope.
9	c	$\beta = \Delta f/f_m = 10/2 = 5$.
10	c	$B = 2(75+15) = 180$ kHz.
11	d	FM amplitude constant ⇒ power independent of β.
12	b	Boost highs before modulation; de-emphasis at Rx cuts them with the noise.
13	c	Ratio detector's large capacitor stabilises the sum voltage — inherent limiting.
14	c	Integrator + balanced (phase) modulator on a crystal carrier → NBFM → multipliers.
15	c	$f_{im} = f_s + 2f_{IF} = 700 + 910 = 1610$ kHz.
16	c	AM: 455 kHz; FM: 10.7 MHz; TV vision: 38.9 MHz.
17	c	Nyquist rate = 2 × 6 = 12 kHz.
18	c	Position (timing) — PPM is a PTM scheme.
19	b	SQNR = 6.02n + 1.76 dB ⇒ ~6 dB/bit.
20	b	Overload when $\Delta \cdot f_s <$ slope; raise Δ or f_s (or use ADM).
21	c	2 bits/symbol ⇒ half the symbol rate ⇒ half the bandwidth.
22	c	Gaussian filter → Gaussian MSK; compact spectrum.
23	b	E1 = 32 × 64 kbps = 2.048 Mbps (T1 = 1.544 Mbps).
24	b	HF refracts off F-layers — multi-hop worldwide coverage.
25	b	$f_c = 9\sqrt{N} = 9 \times 1.1 \times 10^6 = 9.9$ MHz.

#	Ans	Explanation
26	b	$Z_0 = \sqrt{(L/C)} = \sqrt{(0.5\times10^{-6}/200\times10^{-12})} = \sqrt{2500} = 50\ \Omega$.
27	b	Shorted λ/4 inverts to an open (parallel-resonant "metallic insulator").
28	b	$\theta_a = \sin^{-1} 0.242 \approx 14°$.
29	c	35,786 km altitude (radius 42,164 km from earth's centre).
30	b	$D/R = \sqrt{(3N)} = \sqrt{21} \approx 4.58$.

7.20 Rapid-Revision Formula Sheet

Noise

- $P_n = kTB$; $V_n = \sqrt{(4kTRB)}$; $F = SNR_{in}/SNR_{out}$; $T_e = (F-1)\cdot290$
- Friis: $F = F_1 + (F_2-1)/G_1 + (F_3-1)/(G_1G_2)$

Fourier symmetry

- real-even → real-even · real-odd → imaginary-odd · real → $X(-f)=X^*(f)$

AM

- $m = A_m/A_c = (V_{max}-V_{min})/(V_{max}+V_{min})$; $m_t = \sqrt{\Sigma m_i^2}$
- $B = 2f_m$; $P_t = P_c(1+m^2/2)$; **each sideband = $m^2P_c/4$**; $I_t = I_c\sqrt{(1+m^2/2)}$
- $\eta = m^2/(2+m^2)$ → 33.3% max; SSB bandwidth = f_m

FM/PM

- $\beta = \Delta f/f_m$ (FM); $\Delta f = k_f A_m$; Carson: **$B = 2(\Delta f+f_m)$**
- FM power constant = $A_c^2/2R$; broadcast: Δf = 75 kHz, B_{ch} = 200 kHz; pre-emphasis τ = 75 µs
- PLL-FM: output = error/loop-filter voltage; PLL-AM: VCO = coherent carrier → multiplier + LPF

Receiver

- $f_{LO} = f_s + f_{IF}$; **$f_{image} = f_s + 2f_{IF}$**; IFRR = $\sqrt{(1+Q^2\rho^2)}$; IF: 455 kHz (AM), 10.7 MHz (FM)

Pulse & PCM

- $f_s \geq 2f_m$; PWM = PDM = PLM ∈ PTM; PPM demod = ref-pulse + flip-flop → PWM → LPF
- $L = 2^n$; $\Delta = V_{pp}/L$; $N_q = \Delta^2/12$; **SQNR = 6.02n + 1.76 dB**; $R_b = n\cdot f_s$; $B_{min} = R_b/2$
- Voice: 8 kHz × 8 bit = 64 kbps; DM overload: $\Delta f_s \geq 2\pi f_m A_m$

Digital modulation

- QPSK phases **0/90/180/270°**; QPSK BW = ½ BPSK BW; pulse-shaping: sinc, raised-cosine ($B = (1+\alpha)R_s/2$), RRC, Gaussian (GMSK)

Multiplexing

- T1 = 1.544 Mbps (24 ch); **E1 = 2.048 Mbps (30+2)**; FDM group = 12 ch

Propagation

- f_c = 9√N_max; MUF = f_c sec θ; d_LOS(km) = 4.12(√h_t+√h_r)

Transmission lines

- $Z_0 = \sqrt{(L/C)}$; $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$; $S = (1+|\Gamma|)/(1-|\Gamma|)$
- λ/4: $Z_{in} = Z_0^2/Z_L$ (match section $Z_0' = \sqrt{(Z_0 Z_L)}$); λ/2: $Z_{in} = Z_L$
- Reflected power = $|\Gamma|^2$; distortionless: R/L = G/C

OFC

- **$NA = \sqrt{(n_1^2 - n_2^2)} = n_1\sqrt{(2\Delta)}$**; $\theta_a = \sin^{-1}NA$; $V = (2\pi a/\lambda)NA$; single-mode V < 2.405
- Windows: 850/1310/1550 nm; loss ≈ 0.2 dB/km @1550; Rayleigh ∝ $1/\lambda^4$

PA systems

- Constant-voltage line 70.7 V / **100 V**; speaker transformers in parallel; $Z = V^2/P$

Radar

- **$R = c\Delta t/2$** (1 μs ↔ 150 m); $R_{unamb} = c/2PRF$; $\Delta R = c\tau/2$
- R_max = [P_tG²λ²σ/((4π)³S_min)]^¼ ⇒ R ∝ P_t^¼; $f_d = 2v_r/\lambda$

Cellular

- Cluster $N = i^2 + ij + j^2$; reuse 1/N; **$D/R = \sqrt{(3N)}$**; soft handoff = CDMA
- 1G analog → 2G GSM/SMS → 3G WCDMA → 4G LTE/OFDM → 5G NR/mm-wave

Satellite

- GEO height **35,786 km**, period = sidereal day, equatorial; delay ≈ 250 ms/hop
- Uplink > downlink: **C 6/4 GHz, Ku 14/12 GHz**; transponder BW 36 MHz; VSAT dish 0.75–2.4 m, star/hub

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End of Chapter 7. Next: Chapter 8 — Measurement & Instrumentation (as per syllabus sequence).